

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15C.1

CARBOXYLIC ACIDS AND THEIR PHYSICAL PROPERTIES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15C.1: Carboxylic Acids and Their Physical Properties
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15C.1

Q1. 1. (a) Ethanoic acid forms dimers in the gas phase and non-polar solvents. Draw a diagram showing the structure of an ethanoic acid dimer, including hydrogen bonds.

[2 Marks]

WCH14/01/Oct21/Q4(a)

Q2. (b) Explain why ethanoic acid has a higher boiling point (118°C) than propan-1-ol (97°C) despite having similar molar mass.

[3 Marks]

WCH14/01/Oct21/Q4(b)

SECTION B: WORKED MARK SCHEME — 15C.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Ethanoic acid forms dimers in the gas phase and non-polar solvents. Draw a diagram showing the structure of an ethanoic acid dimer, including hydrogen bonds. 1. Two ethanoic acid molecules oriented anti-parallel. 2. Two hydrogen bonds shown between C=O of one molecule and O-H of the other (dotted lines O...H).	[2]
Q2	Q2: (b) Explain why ethanoic acid has a higher boiling point (118°C) than propan-1-ol (97°C) despite having similar molar mass. 1. Both form hydrogen bonds. 2. Ethanoic acid forms dimers via 2 hydrogen bonds per dimer unit. 3. Effectively doubles the molecular mass and strength of intermolecular attraction → higher boiling point.	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15C.1**FAQ #1: What is a carboxylic acid dimer?****Category: Intermolecular Bonding****COMMON EXAMINER TRAP / MISTAKE:**

Drawing only 1 hydrogen bond between two carboxylic acid molecules.

MODEL ANSWER / EXAMINER EXPECTATION:

A dimer consists of TWO carboxylic acid molecules joined by TWO hydrogen bonds between $\text{C}=\text{O}\cdots\text{H}-\text{O}$ of adjacent molecules, forming a 6-membered ring structure.